Molarity and Solution Stoichiometry

Answer key with worked solutions

A. Understanding

1. Molarity.

> Moles of solute per litre of **solution**; unit mol/L (M).

2. Volume of water vs volume of solution.

> Dissolving the solid changes the total volume, so the solution is not exactly 250.0 mL. Dissolve the solid in less water, then add water **up to** a final volume of 250.0 mL (for example in a volumetric flask).

B. Calculating molarity and amounts

3. 10.0 g NaOH in 500.0 mL.

> n = 10.0 g ÷ 40.00 g/mol = 0.250 mol. Molarity = 0.250 mol ÷ 0.5000 L = **0.500 M** (mol/L)

4. KCl for 250.0 mL of 0.150 M.

> n = 0.150 mol/L × 0.2500 L = 0.03750 mol. M(KCl) = 74.55 g/mol, so mass = 0.03750 mol × 74.55 g/mol = **2.80 g**

5. Moles in 35.0 mL of 0.200 M.

> n = 0.200 mol/L × 0.0350 L = **0.00700 mol**

C. Dilution

6. Volume of 12.0 M HCl needed.

> $V_1 = \dfrac{M_2 V_2}{M_1} = \dfrac{(0.600\ \text{M})(500.0\ \text{mL})}{12.0\ \text{M}} =$ **25.0 mL** (the units of M cancel)

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7. New concentration.

> $M_2 = \dfrac{M_1 V_1}{V_2} = \dfrac{(2.00\ \text{M})(50.0\ \text{mL})}{400.0\ \text{mL}} =$ **0.250 M** (the units of mL cancel)

D. Titrations

8. Concentration of HCl.

> n(NaOH) = 0.120 mol/L × 0.02250 L = 0.002700 mol = n(HCl) (1 : 1). [HCl] = 0.002700 mol ÷ 0.02500 L = **0.108 M**

9. Concentration of H_2SO_4.

> n(NaOH) = 0.200 mol/L × 0.01875 L = 0.003750 mol. n(H_2SO_4) = 0.003750 mol NaOH × (1 mol H_2SO_4 / 2 mol NaOH) = 0.001875 mol. [H_2SO_4] = 0.001875 mol ÷ 0.01500 L = **0.125 M**

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